



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Aviation MRO and Software Supply Chain Cyber Risk Require... Detection Package

Detection Engineering Package

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Aviation MRO and Software Supply Chain Detection Package

1. Executive Summary

This detection engineering package operationalizes `LBC-TR-2026-134`, which identified aviation maintenance, repair, overhaul, and software supply-chain risk paths that can expose aviation operators through trusted suppliers, updater channels, remote access paths, and shared service dependencies.

The package provides one deployable Sigma analytic, one Microsoft Defender/Sentinel KQL analytic, one YARA hunt rule, one Suricata syntax-validated network rule, and one broader KQL hunt. The first validation pass is static/package validation: syntax and package completeness checks pass, while live SIEM execution remains planned for Phase 2 lab replay.

2. Source Intelligence

Field	Value
Source report	`LBC-TR-2026-134`
Source title	Aviation MRO and Software Supply Chain Cyber Risk Requires Third-Party Assurance
Source path	`reports/threat-reports/LBC-TR-2026-134_aviation-mro-and-software-supply-chain-cyber-risk-requires-third-party-assurance`
ATT&CK coverage	T1190, T1059, T1567
Primary defender question	Which trusted supplier, updater, edge, or remote-access behaviors should aviation defenders monitor first?

3. Behavior Coverage Matrix

Behavior	ATT&CK	Telemetry required	Detection files	Hunt files	Status
Suspicious script, LOLBin, archive, or transfer tool execution from trusted updater/VPN/browser/remote/service parents	T1059	Endpoint process telemetry	`detections/sigma/trusted-channel-suspicious-child-process.yml`, `detections/kql/trusted-channel-suspicious-child-process.kql`		draft; static validation passed
Abnormal outbound activity from supplier, updater, VPN, or remote-access tooling	T1567	Endpoint network, DNS, proxy, firewall, or Zeek telemetry	`detections/suricata/aviation-supplier-unusual-http.rules`	`hunts/kql/supplier-remote-access-network-anomaly.kql`	draft; Suricata syntax validation passed
Suspicious script/download/archive artifacts discovered during supplier-channel triage	T1059	File, script, and collected artifact scanning	`detections/yara/aviation_supply_chain_suspicious_scripts.yar`		draft; YARA compile validation passed

Public-facing or supplier-managed service exposure requiring telemetry review	T1190	Edge service, vulnerability, authentication, and remote access logs		covered as triage pivot in hunt	planned
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4. Detection Package

4.1. Sigma

`detections/sigma/trusted-channel-suspicious-child-process.yml` detects suspicious child processes spawned by likely trusted-channel parents such as updater, VPN, browser, remote access, MSP, or service-management binaries.

Deployment notes:

- Tune parent process terms for known aviation supplier tooling, EFB support agents, MRO portals, VPN clients, and remote management tools.
- Baseline software deployment systems before alerting at high severity.
- Prioritize hits involving encoded PowerShell, archive creation, credential access, download utilities, or unusual user context.

4.2. Microsoft Defender / Sentinel KQL

`detections/kql/trusted-channel-suspicious-child-process.kql` implements the same parent-child process logic in Defender XDR / Sentinel style KQL.

4.3. YARA

`detections/yara/aviation\_supply\_chain\_suspicious\_scripts.yar` provides a lightweight artifact-scanning hunt for suspicious script/download/archive strings encountered during trusted supplier-channel triage.

4.4. Suricata

`detections/suricata/aviation-supplier-unusual-http.rules` provides a starter network analytic for curl-style HTTP transfer behavior. Tune this heavily before production deployment because user-agent based network content is broad.

4.5. Hunt Query

`hunts/kql/supplier-remote-access-network-anomaly.kql` hunts for unusually high outbound connection volume or endpoint diversity from supplier-access and trusted-channel processes.

5. Validation Plan

Analytic	Fixture / test data	Expected result	Current result	Notes
Sigma trusted-channel suspicious child process	`validation/fixtures/process-events.jsonl`	Match updater/VPN parent launching shell/script process	static/package validation passed	Live Sigma backend execution not yet wired
Defender KQL trusted-channel suspicious child process	`validation/fixtures/process-events.jsonl`	Match trusted-channel parent launching suspicious child process	static/package validation passed	Live KQL replay not yet wired
Supplier remote-access network anomaly hunt	`validation/fixtures/process-events.jsonl`	Produce reviewable aggregate when network fixture exists	structural validation passed	Needs richer network fixture in next iteration
YARA suspicious script artifact rule	`validation/fixtures/suspicious-script.txt`	Compile with `yara`; fixture scan should match	backend compile validation passed; YARA fixture scan	Fixture contains suspicious PowerShell/download

			passed	strings
Suricata unusual supplier HTTP rule	`validation/fixtures/curl-http-transfer.pcap`	`suricata -T` syntax check passes and PCAP replay produces an alert	backend syntax validation passed; PCAP replay passed	Fixture contains a curl-style HTTP transfer over an established TCP flow

6. False Positives and Tuning

Expected benign lookalikes include software update scripts, endpoint management tools, vulnerability scanners, remote support tools, packaging jobs, and MSP maintenance windows. Suppression should be based on known-good signer, known-good parent path, expected maintenance window, and managed deployment host role rather than only filename.

7. Blind Spots

This package does not prove exploitation or compromise by itself. It requires endpoint process telemetry for the primary analytic and endpoint/network telemetry for the hunt. Environments without command-line logging, parent-process capture, remote URL capture, or supplier access inventory will need additional logging before deployment.

8. Analyst Handoff

1. Review parent-process terms against the aviation operator's approved supplier, VPN, MRO, EFB, updater, and MSP tooling.
2. Deploy the Sigma or KQL analytic in monitor mode.
3. Run the hunt across 14 to 30 days of process/network telemetry.
4. Escalate hits that combine suspicious child execution, unusual outbound connections, archive creation, credential access, or supplier remote-access anomalies.
5. Record live validation results under `validation/results/` before promoting `detection\_status` beyond draft. Current backend validation covers Sigma/KQL structural checks plus local `yara` and `suricata -T` execution.